8 **(a)** probability of decay (of a nucleus)/fraction of number of nuclei in sample that decay M1
per unit time A1 [2]
(*allow λ =(dN / dt) / N with symbols explained – (M1), (A1)*)

(b) **(i)** number $= (1.2 \times 6.02 \times 10^{23}) / 235$ C1
$= 3.1 \times 10^{21}$ A1 [2]

(ii)	$N = N_0\, e^{-\lambda t}$		
	negligible activity from the krypton	B1	
	for barium, $N = (3.1 \times 10^{21}) \exp(-6.4 \times 10^{-4} \times 3600)$		
	$= 3.1 \times 10^{20}$	C1	
	activity $= \lambda N$		
	$= 6.4 \times 10^{-4} \times 3.1 \times 10^{20}$	C1	
	$= 2.0 \times 10^{17}$ Bq	A1	[4]

Section B